

FitSteps: Micro Activity Tracking System

1 Problem Statement

Consistent physical activity tracking remains fragmented for users who do not own dedicated wearable hardware such as smartwatches or fitness bands. Existing commercial fitness applications predominantly rely on background sensor telemetry, continuous GPS monitoring, or proprietary wearable ecosystems. This hardware reliance introduces significant friction for casual users, students, and individuals seeking straightforward movement logging.

Furthermore, complex mobile fitness suites often overwhelm users with excessive metrics, social features, and restrictive paywalls. Without a lightweight and accessible tool, individuals struggle to maintain visibility over daily activity trends and personal fitness goals.

2 Solution Overview

FitSteps is a micro activity tracking system designed to eliminate hardware dependency by providing a simple web-based platform for manual activity logging and goal tracking. Users can record activities based on duration, distance, calories burned, and activity type.

The system converts these inputs into structured analytics such as weekly summaries, activity distribution, and consistency metrics. This reduces user friction while enabling continuous awareness of movement patterns.

3 System Architecture

FitSteps follows a decoupled client-server architecture using RESTful communication.

React (Vite) Frontend <--> Axios HTTP <--> FastAPI Backend <--> SQLite Database

The frontend handles user interaction and visualization, while the backend processes business logic, validation, and database operations.

4 Core Functional Modules

- **Activity Tracking:** Provides CRUD operations for logging workouts, including duration, distance, calories, category, and timestamp.
- **Goal Management:** Enables users to define daily or activity-based goals with measurable targets and track progress over time.
- **Analytics Engine:** Computes weekly averages, identifies best-performing days, calculates goal completion rates, and tracks user activity streaks.
- **Activity Categorization:** Classifies activities into walking, running, cycling, and workouts for aggregated analysis.

5 Analytics Design

The analytics system uses deterministic logic based on stored activity data:

- **Step Estimation:** Distance-based activities use the formula:

$$steps = distance(km) \times 1300$$

Duration-based activities use:

$$steps = duration(minutes) \times 100$$

- **Weekly Average:** Total estimated steps over the last 7 days divided by 7.
- **Best Day Detection:** Identifies the day with the highest aggregated activity based on calories or duration.
- **Goal Completion Rate:** Percentage of completed goals relative to total goals.
- **Streak Calculation:** Counts consecutive days with at least one recorded activity.

6 Deployment Architecture

The system is deployed using separate services for frontend and backend:

- **Frontend (Vercel):** Hosts optimized static assets on a global CDN for fast delivery.
- **Backend (Render):** Runs the FastAPI application in a managed environment using Uvicorn.

7 Limitations

- No integration with wearable devices or health APIs.
- Step calculations are estimations, not sensor-based measurements.
- No authentication or multi-user support.

8 Conclusion

FitSteps demonstrates that effective activity tracking can be achieved through a lightweight and focused system design. By prioritizing usability and essential analytics over complex integrations, the system enables consistent habit tracking with minimal user effort.